Proposal: Replace the sourcing reverse block scan with an index lookup

Status	Proposal
Area	JetStream stream sourcing (`server/stream.go`) + store index (`server/filestore.go` , `server/memstore.go`)
Date	2026-06-06
Companion	`sourcing-durable-resume.md` (lifecycle, durable consumer, triggers)
Target	`startingSequenceForSources` (`stream.go:4694`) and `setStartingSequenceForSources` (`stream.go:4566`)

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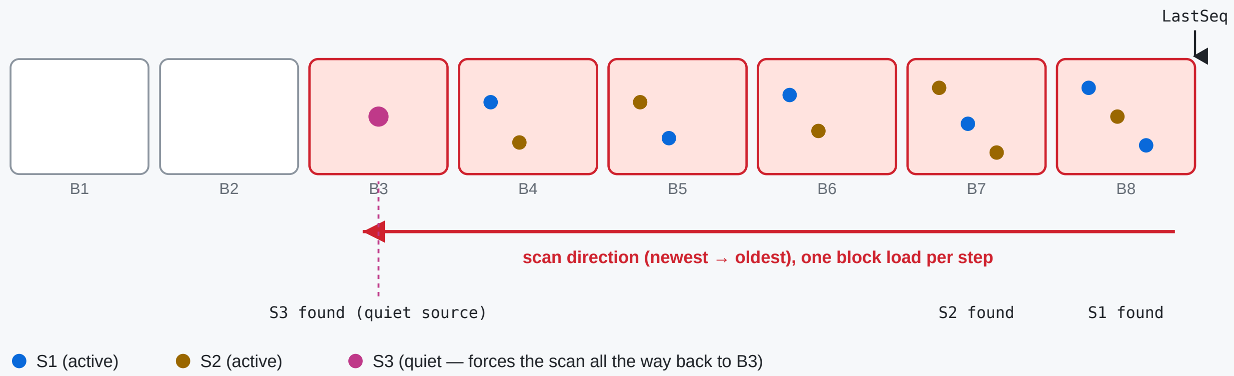
1. The problem in one paragraph

On hub-stream **leader election / restart**, `setupSourceConsumers` calls `startingSequenceForSources` **unconditionally** (`stream.go:4821`). That function reconstructs each source's last sourced sequence by walking the stream's **own store backwards from `LastSeq`**, one block at a time (`LoadPrevMsgMulti` , `filestore.go:9425`), reading the `Nats-Stream-Source` header off matching messages, until **every** source has been located.

> Note: the walk is not as naive as message-by-message — within each block the per-block index (`fss`) skips non-matching messages, and the per-source sublist narrows as sources are found. But `LoadPrevMsgMulti` still **visits every block** from `LastSeq` back to the match (loading/decompressing cold ones), and the outer loop **re-walks** as the sublist narrows. So the cost grows with both the number of blocks back to the least-recently-active source (worst case the **entire store**) and the number of sources — all under `mset.mu` .

The reverse block scan: rebuild each source's last sequence from stored headers

startingSequenceForSources walks back from LastSeq, LoadPrevMsgMulti loads (+decompresses) a block at a time until every source is found.



Why it can be slow on a limits-based stream

A single quiet/sparse source (S3) forces the scan past every newer block — worst case the entire stream. Each uncached block is read from disk, decrypted and decompressed. With many sources and a large store, leader election stalls under mset.mu while this runs.

2. The key insight

The store **already maintains a per-subject index** that can return the last sequence for a subject *without* walking messages:

- `fs.psim` — per-subject info, including `lblk` = the **last block index** that contains each subject (`filestore.go:3843-3852` , `8980-8984`).
- `mb.fss` — per-block subject state with `Last` per subject (`filestore.go:9003-9015`).

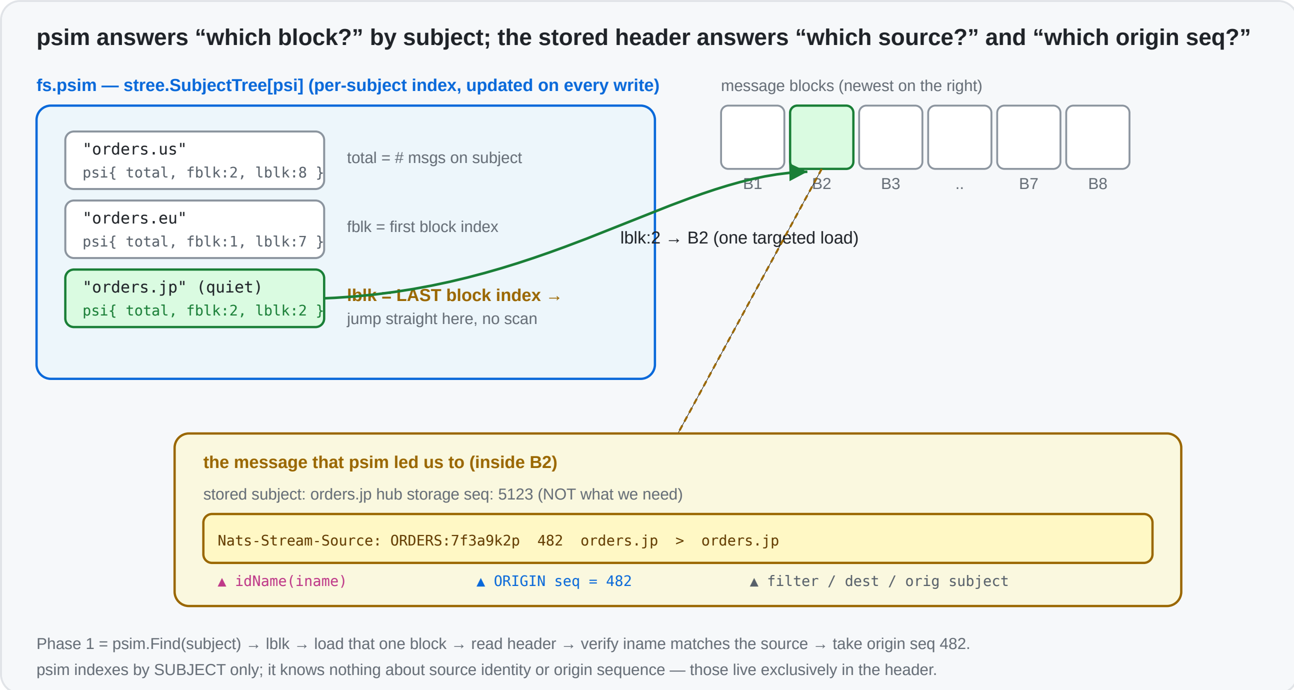
This is what powers `LoadLastMsg(subject)` → `loadLast` (`filestore.go:8939`): for a literal subject it jumps straight to `info.lblk` and reads that subject's last message — **one targeted block load**, regardless of how far back it is. `MultiLastSeqs` (`filestore.go:3806`) does the same for a batch of filters in a single index pass.

The current scan ignores this index and instead does a generic backward message walk. The header it needs (the **origin** sequence) is right there in the message that `LoadLastMsg` already returns.

> Why a plain "last seq for subject" isn't enough on its own: `si.sseq` must be the **origin** stream sequence taken from the `Nats-Stream-Source` header, not the hub's storage sequence. So we use the index to *find* the source's last stored message in O(1), then read the origin seq from its header.

3. Background: the subject index (`psim` + `fss`) and the source header

The filestore keeps a **two-level subject index**. Phase 1 stands entirely on it, so it's worth being precise about what each level is.



3.1 `psim` — the store-wide "which blocks hold this subject" map

`fs.psim` (`filestore.go:195`) is a `stree.SubjectTree[psi]` — a subject-token tree keyed by the full message subject — whose value is a tiny per-subject record (`filestore.go:168`):

```
type psi struct {
    total uint64  // how many messages exist for this subject across the whole s
    fblk  uint32  // index of the FIRST block that still contains this subject
    lblk  uint32  // index of the LAST  block that still contains this subject
}
```

It is maintained incrementally on every write (`filestore.go:4933-4943`): on store, the subject's entry is created (`total:1, fblk=lblk=current block`) or its `total` is bumped and `lblk` advanced to the current block; on removal/expiry the counts and `fblk` / `lblk` are walked forward as blocks empty out. Because `psim` is a *subject tree*, it supports both exact `Find(subject)` and wildcard `Match(filter)` in time proportional to the subject token structure — **not** to the number of messages.

The single field that makes Phase 1 cheap is `lblk`: given a source's subject, `psim.Find` returns the **exact last block** that contains it. There is no need to scan newer blocks — most of which don't contain that subject at all. (`MultiLastSeqs`, `filestore.go:3806`, uses the same `psim` to answer a *batch* of filters in one index pass.)

3.2 `fss` — the per-block "first/last seq of this subject in this block" map

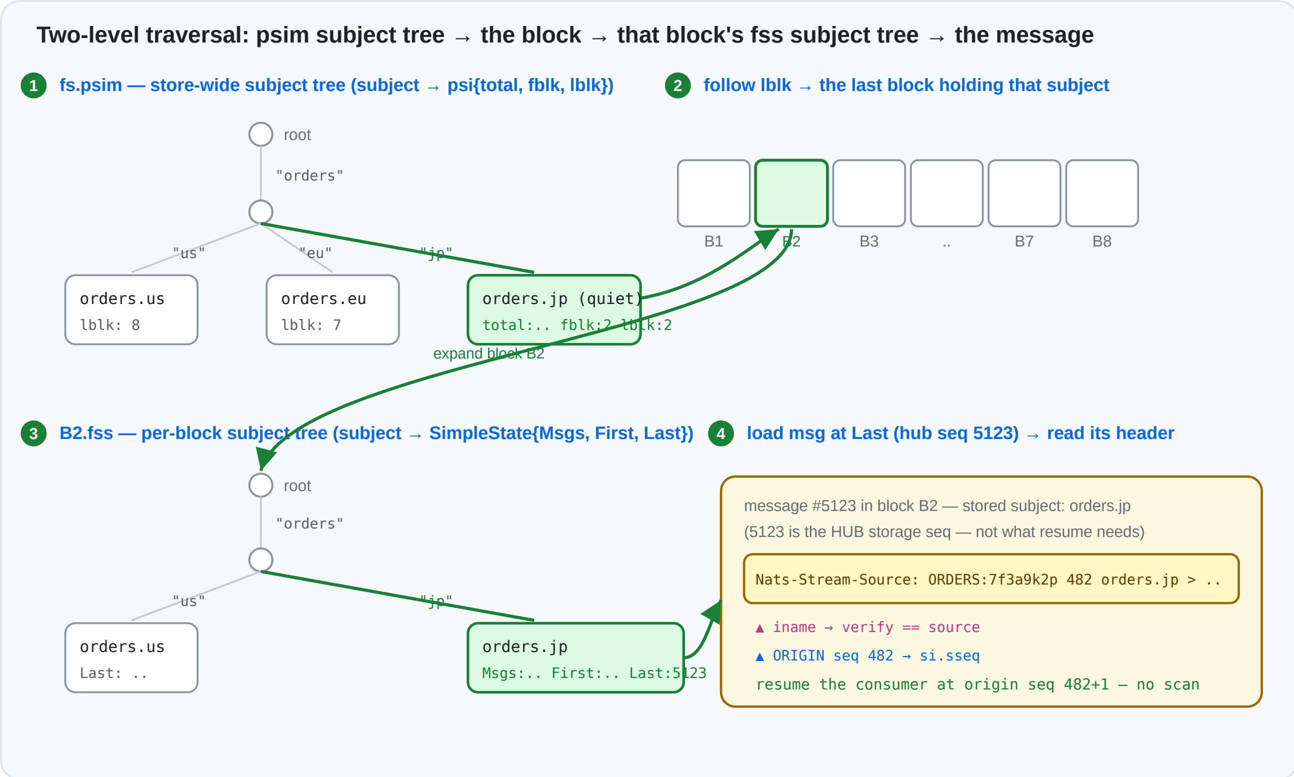
`psim` gets us to a block; `mb.fss` (`filestore.go:239`) finishes the job *inside* that block. Each `msgBlock` carries its own `stree.SubjectTree[SimpleState]` mapping subject → (`store.go:180`):

```
type SimpleState struct {
    Msgs  uint64 // messages for this subject in THIS block
    First uint64 // first sequence for this subject in this block
    Last  uint64 // last  sequence for this subject in this block
    // (firstNeedsUpdate / lastNeedsUpdate: lazily recomputed when interior dele
}
```

So the two levels compose into an O(1)-ish lookup, which is exactly what `loadLast` (`filestore.go:8980-9034`) does:

```
psim.Find(subject) ─► info.lblk ─► block ─► block.fss.Find(subject) ─► ss.Last ─
```

`fss` is what gives us the precise sequence within the block (and it correctly skips interior-deleted messages via `lastNeedsUpdate` / `recalculateForSubj`). On a cold block, `ensurePerSubjectInfoLoaded` loads/derives `fss` for that **one** block — the cost we pay once per resolved source, versus the current scan paying it for every block back to the oldest source.



3.3 Why each message carries a `Nats-Stream-Source` header

`psim` / `fss` index by **subject**. They know nothing about *which source* produced a stored message or *where in the origin stream* it came from — and those are precisely the two facts resume needs. That information is carried only in the per-message header `Nats-Stream-Source` (`JSStreamSource`, `stream.go:635`), written by `genSourceHeader` (`stream.go:4470`) as:

```
Nats-Stream-Source:  <idName>  <originSeq>  <filter>  <destination>  <origSubjec
                       |          |
                       |          └─ the ORIGIN stream's sequence — what we must
                       └─ origin stream name (+ domain/consumer hash); with filt
                          reconstructs the source's full iname  (parsed by strea
```

This header is needed for three independent reasons:

1. **Source attribution.** A hub message is stored under its *destination* subject; the storage layer records nothing about its origin. When two sources can land on the same/overlapping subject, only the header's `iname` says which source a given message belongs to — hence Phase 1's `iname == si.iname` check.
2. **Origin-sequence tracking.** The sourcing consumer is (re)created on the **origin** with `DeliverByStartSequence`, which lives in the *origin's* sequence space. The hub's own storage sequence (e.g. 5123) is unrelated to the origin sequence (e.g. 482). Only the

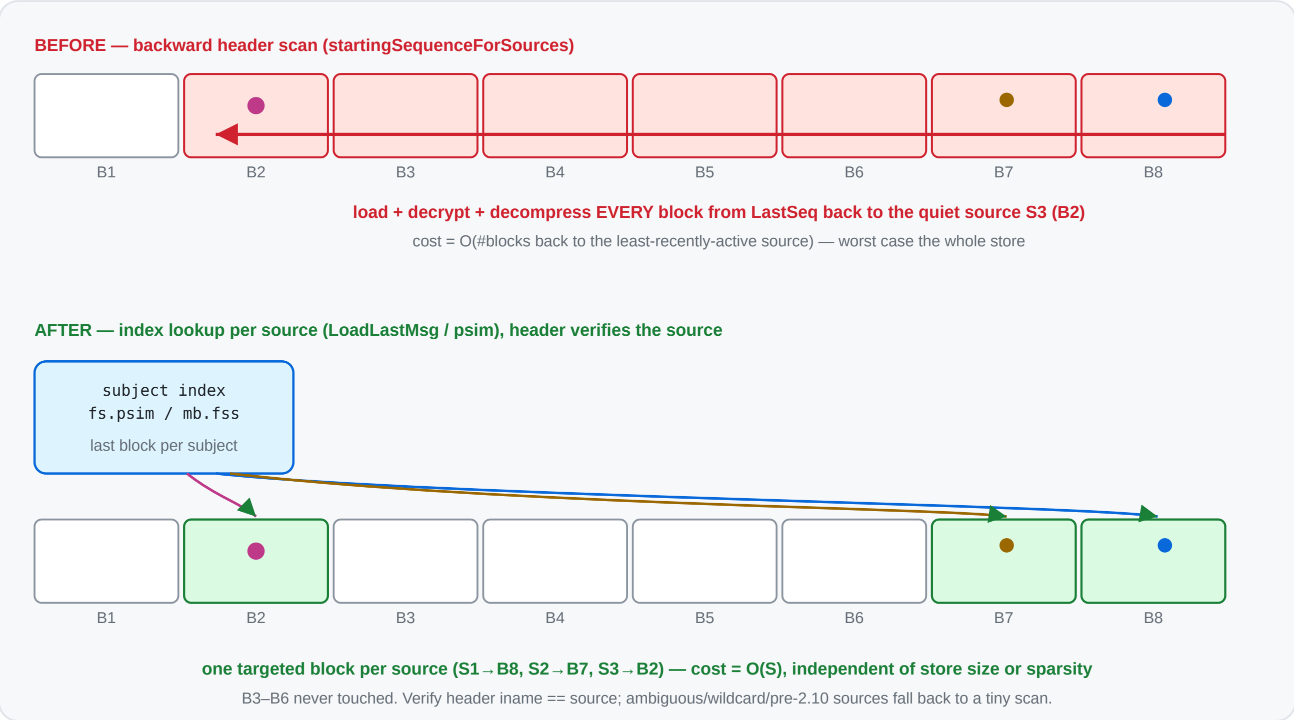
header preserves the origin sequence, so it is the *only* place a resume point can be recovered from without contacting the origin.

3. **Loop / daisy-chain handling & dedup.** When re-sourcing (A→B→C), `processInboundSourceMsg` strips the inbound `JSStreamSource` and stamps its own (`stream.go:4398` , `4405`), so each hop's provenance is well-defined and re-sourced duplicates can be recognised.

This is the crux of the whole proposal: **the index tells us *which block* in O(1); the header (read from that one message) tells us *which source* and *which origin seq*.** Today's scan reads the header off *every* message on the way back because it never consults `psim` / `fss` to jump directly to the right one.

4. Proposed algorithm

Two phases. Phase 1 resolves the common case with index lookups; Phase 2 keeps today's scan only for sources that genuinely can't be attributed by subject.



For each source, its **stored subject** is the `FilterSubject` for a plain source, or — for a source with a **single, concrete (non-templated)** subject transform — the transform **Destination**. Templated transform destinations contain `{{...}}` mapping tokens (the rendered subject varies per message), and partial wildcards in a destination are rejected at config time, so a destination is concrete unless it is empty, `>`, or contains `{{`.

```
Phase 1 — index lookup (per source, O(1) targeted load):
  for each source si:
```

```
      subj := si.FilterSubject                     // plain source
      if si has exactly one transform: subj := transform.Destination
      if subj == "" || subj == ">" || subj contains "{{":  defer to Phase 2; con
      sm := store.LoadLastMsg(subj)                // jumps via psim, ~1 block
      if sm has a JSStreamSource header:
          (_, iname, sseq) := streamAndSeq(header)
          if iname == si.iname:                    // self-verifying attribution
              si.sseq = sseq; mark resolved
  // anything not resolved (no header / header for a different source / wildcard
  //  multi-or-templated transform / pre-2.10) → Phase 2

Phase 2 — fallback reverse scan (only for the unresolved subset):
  run today's LoadPrevMsgMulti loop, but with the sublist built from ONLY the
  unresolved sources, so it narrows immediately and stops as soon as they are fo
```

The header check makes Phase 1 **self-correcting**: if a subject is shared between sources, or overlaps a direct publish on the hub, or the last message has no source header, the lookup simply doesn't match and that source drops to Phase 2 — never producing a wrong answer.

5. Before → after (code)

Before (`startingSequenceForSources`, condensed — `stream.go:4694`)

```
mset.resetSourceInfo()
// build a sublist of ALL sources' subjects
refreshSublist()                                   // every source
for last := state.LastSeq; ; {                     // walk the WHOLE store backw
    sm, seq, err := mset.store.LoadPrevMsgMulti(sl, last, &smv) // loads/decompr
    if err == ErrStoreEOF || err != nil { break }
    last = seq - 1
    ...
    _, iName, sseq := streamAndSeq(bytesToString(ss))
    update(iName, sseq)                            // narrows sublist as sources
    if len(seqs) == expected { return }            // stop only when ALL sources
}
```

Cost: O(blocks back to the least-recently-active source) block loads → worst case the entire store.

After

```
mset.resetSourceInfo()
var smv StoreMsg
unresolved := map[string]*sourceInfo{}

// Phase 1: targeted index lookup per source.
for _, ssi := range mset.cfg.Sources {
    si := mset.sources[ssi.iname]
    if si == nil { continue }
    var subj string                                // FilterSubject, or single c
    switch {
    case len(ssi.SubjectTransforms) == 0: subj = ssi.FilterSubject
    case len(ssi.SubjectTransforms) == 1: subj = ssi.SubjectTransforms[0].Destin
    }
    if subj == _EMPTY_ || subj == fwcs || strings.Contains(subj, "{{") {
        unresolved[ssi.iname] = si                 // can't attribute by a singl
        continue
    }
    sm, err := mset.store.LoadLastMsg(subj, &smv)  // O(1) via psim/fss, returns
    if err != nil || sm == nil || len(sm.hdr) == 0 {
        unresolved[ssi.iname] = si
        continue
    }
    if ss := sliceHeader(JSStreamSource, sm.hdr); len(ss) > 0 {
        if _, iName, sseq := streamAndSeq(bytesToString(ss)); iName == ssi.iname
            si.sseq, si.dseq = sseq, 0             // verified — resolved withou
            continue
        }
    }
    unresolved[ssi.iname] = si
}

// Phase 2: only the leftovers pay the (now tiny) reverse scan.
if len(unresolved) > 0 {
    mset.reverseScanForSources(unresolved)         // = today's loop, scoped to
}
```

Cost: Phase 1 = O(S) targeted lookups (recent sources share/cache the last block; a quiet source is **one** targeted load, not a walk). Phase 2 only runs for ambiguous sources, with a smaller sublist.

> Multi-destination transform sources: call `LoadLastMsg` for each destination and keep the highest origin `sseq`, or defer them to Phase 2. Either is fine; deferring is simplest to start.

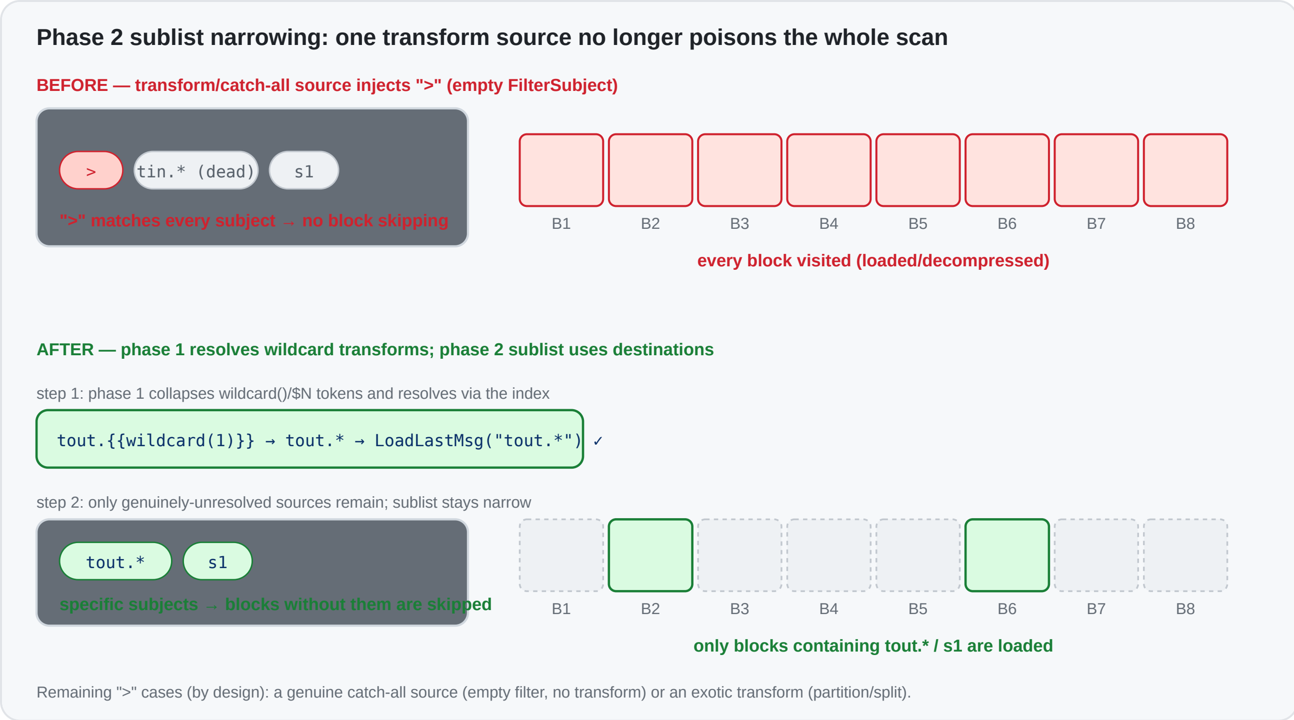
6. Before → after (complexity)

	Before	After
Index used	none (generic message walk)	`psim` / `fss` per-subject last-block index
Blocks loaded	back to least-active source (≤ all blocks)	~1 per source, only the blocks actually holding a source's last msg
Sensitivity to a **quiet** source	drags the scan back across the whole store	none — jumps straight to its block
Sensitivity to **store size**	linear in blocks scanned	none
Worst case still possible?	normal case	only if every source is `>`/wildcard/shared-subject/pre-2.10 (→ Phase 2 == today)
Held lock	`mset.mu` for the whole walk	`mset.mu` for O(S) lookups; Phase 2 only for leftovers

The edge→hub topology (each edge mapped to a distinct subject/domain) lands entirely in Phase 1 → the backward scan disappears for that case.

7. Phase 2: keeping the fallback scan narrow

Phase 1 removes most sources from the picture, but whatever remains still runs the reverse `LoadPrevMsgMulti` scan. That scan's whole speed advantage depends on the **sublist being narrow** — `prevMatchingMulti` uses each block's `fss` to skip blocks that contain none of the sublist's subjects. A single `>` (full wildcard) in the sublist defeats that: it matches every subject, so no block can be skipped and the scan visits every block back to the match.



What used to inject `>`

1. **Every transform source.** A transform source has an empty `FilterSubject` (mutually exclusive with `SubjectTransforms`, `stream.go:1976`), so the old `refreshSublist` did `sl.Insert(fwcs)`. One transform source in the unresolved set widened the whole sublist to `>` → full backward scan.
2. **Templated transforms were deferred to Phase 2 at all.** Their *rendered* destination is a narrow wildcard (e.g. `tout.{{wildcard(1)}}` always lands on `tout.*`), so they don't need a full scan.
3. **Dead entries.** Phase 2 also inserted `si.sfs` — the transform *source* filters (e.g. `tin.*`) — which never match stored subjects (messages are stored under the *destination*).

The two changes

- **Resolve wildcard transforms in Phase 1.** `transformUntokenize(dest)` collapses `wildcard()` / `$N` mapping tokens to a subject wildcard (`tout.{{wildcard(1)}}` → `tout.*`); `LoadLastMsg` accepts that wildcard and the header-`iname` check still guards attribution. So the common templated transforms resolve in Phase 1 and never reach the scan.
- **Build the Phase 2 sublist from destinations.** For any transform source that *does* reach Phase 2 (e.g. its wildcard space is shared with another source, so Phase 1 saw a header for the other one), insert the destination's wildcard form instead of `>`, and drop the dead `si.sfs` inserts.

What still forces `>` (by design)

Only a source that *genuinely* needs a broad match:

- a **catch-all source** — empty `FilterSubject`, no transform — sourcing every subject of its origin; and
- an **exotic transform** (`partition`/`split`/`slice`/…) whose rendered destination isn't a subject wildcard, so `transformUntokenize` can't reduce it (it leaves the `{{…}}` token in place, which we detect and fall back to `>`).

Both are uncommon for the edge→hub fan-in. Distinct subjects, wildcard filters, and `wildcard()`/`$N` transforms all either resolve in Phase 1 or keep the Phase 2 sublist narrow enough to skip blocks.

> Correctness note: this is purely about *narrowing*, never about *missing* a source. If a narrowed subject is wrong (shared space), the header check fails and the source stays in the unresolved set; an exotic/empty case falls back to `>`. So Phase 2 still finds every source it did before — it just loads far fewer blocks getting there.

8. Correctness & edge cases

- **Same semantics.** Today's scan records, per source, the **most recent** stored message's origin seq (first hit scanning backward). `LoadLastMsg` returns exactly that message. Deleted/interior messages are handled by `loadLast` (it skips `dmap` entries and walks to the previous block if needed), matching the scan's behaviour.
- **Shared subjects / direct publishes.** Resolved safely by the header-iname verification → fall to Phase 2.
- **Wildcard / empty (`>`) filters.** Can't be attributed to one source by subject → Phase 2 (same as today for them).
- **Pre-2.10 headers** (no `iname`, only stream name): verification won't match → Phase 2, which keeps the existing stream-name matching path.
- **memstore.** `MemStore` has the simpler linear `LoadPrevMsgMulti`; `LoadLastMsg` there is also index-light, but memstore streams are small, so Phase 2 cost is negligible. (We can add a memstore `loadLast` fast path later if needed.)
- **`setStartingSequenceForSources`** (the `STREAM.UPDATE` path, `stream.go:4566`) gets the same Phase 1 treatment for the subset of sources it processes.

9. Risks

- **Index freshness.** `psim`/`fss` may lazily need a recalculation (`lastNeedsUpdate`, `recalculateForSubj`); `loadLast`/`MultiLastSeqs` already handle that, so we inherit correct behaviour.

- **Behavioural parity.** Phase 1 must produce identical `si.sseq` to the old scan for non-ambiguous sources; this is the core test (below). Keeping Phase 2 byte-for-byte as today bounds the blast radius.
- **Encryption/compression.** Phase 1 still loads the one block holding a source's last message (decrypt/decompress), but only that block — no change in correctness, large reduction in volume.

10. Testing & validation

Prototype status — implemented ✅

A working prototype of the two-phase resolver is implemented in `startingSequenceForSources` (`server/stream.go`), with:

- `TestJetStreamStartingSequenceForSourcesIndexFastPath` (`server/jetstream_sourcing_resume_test.go`) — three distinct-subject sources (index fast path) plus one **subject-transform** source (phase 2 fallback), each with a different origin depth and buried under 20k direct publishes. Asserts every recovered `si.sseq` equals the expected last origin sequence. **Passes.**
- `TestJetStreamStartingSequenceForSourcesAmbiguity` — a catch-all (empty filter) source and **two sources transformed onto the same destination subject** (shared stored subject), mixed with a distinct-subject source. Asserts the phase 2 fallback disambiguates all of them. **Passes.**
- `TestJetStreamSetStartingSequenceForSourcesIndex` — the same fast path applied to the `STREAM.UPDATE` twin `setStartingSequenceForSources` (distinct subject + catch-all + a **concrete-destination transform** source), clearing and recovering `sseq` . **Passes.**
- **Transform fast path:** a source with a single subject transform is resolved in Phase 1 by collapsing `wildcard()` / `$N` mapping tokens to a wildcard form (`transformUntokenize` , e.g. `tout.{{wildcard(1)}}` → `tout.*`) and doing `LoadLastMsg(wildcardForm)` , verified by the header. Only multi-transform and **exotic** transforms (`partition` / `split` /… that don't reduce to a subject wildcard) still defer to Phase 2.
- **Phase 2 no longer degrades to `>` for transform sources.** Previously a transform source (empty `FilterSubject`) inserted a `>` catch-all into the sublist, defeating per-block skipping for the whole scan. The Phase 2 sublist is now built from the transform **destinations** (wildcard form), with the dead `si.sfs` source-filter inserts removed; only a genuine catch-all source or an exotic transform still forces `>` .
- The change was also applied to `setStartingSequenceForSources` itself (`stream.go:4566`), so both the leader-election and the config-update resume paths use the index fast path and the narrowed sublist.

- Tests (all pass, incl. `-race`): `…IndexFastPath`, `…Ambiguity` (catch-all + shared concrete destination), `…TemplatedTransform` (wildcard transform), `…SharedWildcardTransform` (two templated transforms sharing a `gout.*` space), `TestJetStreamSetStartingSequenceForSourcesIndex` (twin), `TestJetStreamSourcingResumeAfterRolloutRestart` (end-to-end hard-restart, exactly-once resume), `TestJetStreamClusterSourcingResumeAfterLeaderStepDown` (R3 leader-election resume), `TestJetStreamStartingSequenceForSourcesDifferential` (randomized layouts vs. a brute-force reference), `…MemStore` (memory storage), `…PartitionTransform` (exotic `>` fallback), `…SeededEdges` (pre-2.10 headers, subject overlap, interior delete), `…ConfigUpdateScoping` (update add/remove scoping), and `…FirstSeqAdvanced` (resume after `FirstSeq` advances).
- `BenchmarkJetStreamScanForSources` (existing, single source), `BenchmarkJetStreamScanForSourcesMulti` (16 sources spread across the store), `BenchmarkJetStreamSourceResumeLeafnodeFanIn` (8–512 edges feeding a hub, quiet edges buried under an all-sourced tail), `…FanInTailDepth` (fixed edges, varying tail depth — isolates the store-depth axis), `…PartitionFallback` (the by-design `>` path), and `…DeepStore` (direct store seeding up to 16/64/256 sources × 10M, for at-scale and cross-design comparison). See the tables below.
- Existing sourcing suite (`SourceBasics`, `SourceRemovalAndReAdd`, `WorkQueueSourceRestart`, `SourceWorkingQueueWithLimit`, `StreamSourceWithoutDuplicateWindow`, `MirrorAndSourcesFilteredConsumers`) still passes.

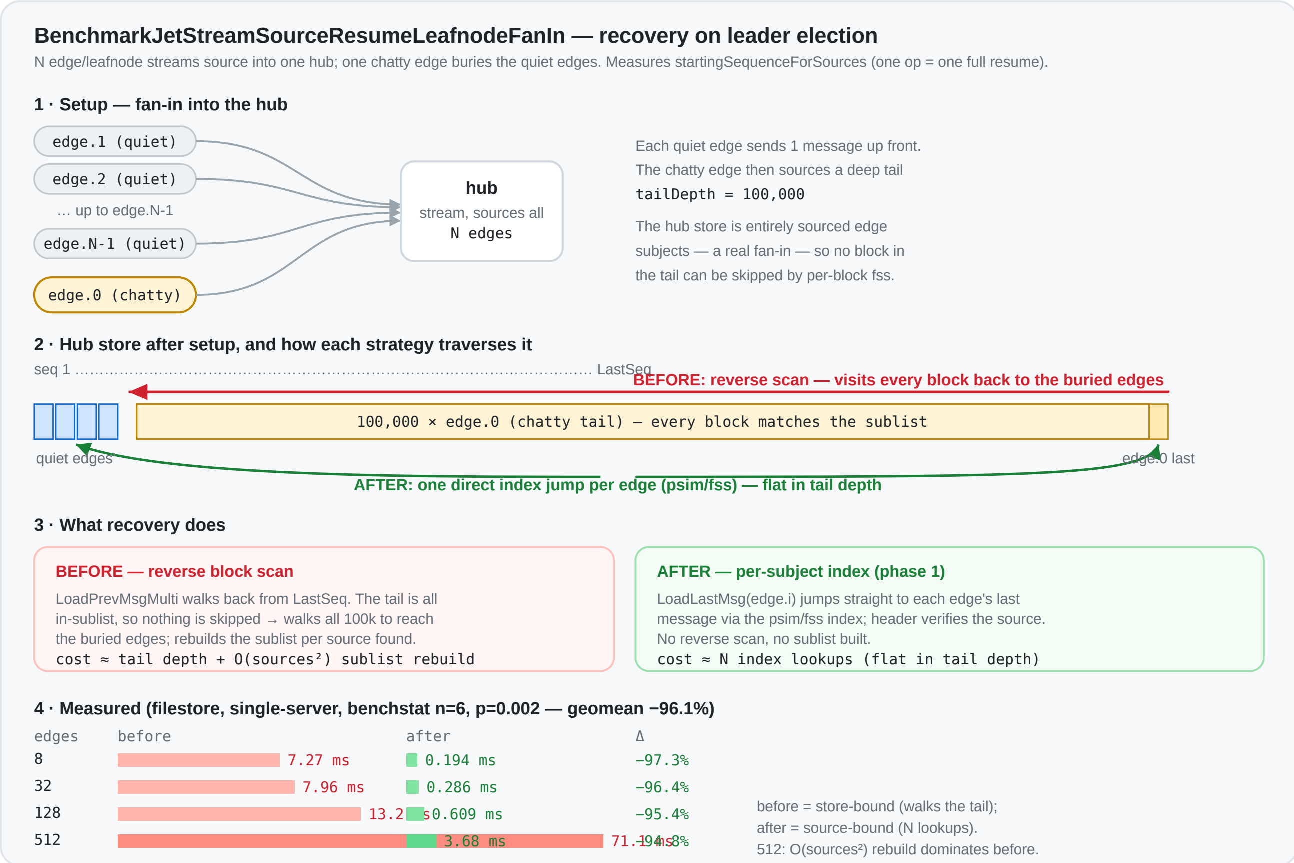
Measured (filestore, single-server)

Benchmark	Before (reverse scan)	After (phase 1)	Speedup
`ScanForSources` — 1 source, ~100k buried msgs	~8.5 µs/op	~5.5 µs/op	~1.5×
`ScanForSourcesMulti` — 16 sources, spread	~110.8 µs/op	~14.0 µs/op	**~7.9×**

The single-source gap is modest because `LoadPrevMsgMulti` already skips non-matching messages per block; the win scales with the number of sources (each avoids a re-walk via a direct `psim` jump), which is exactly the edge→hub fan-in case.

Leafnode fan-in resume (the leader-election case)

`BenchmarkJetStreamSourceResumeLeafnodeFanIn` models the scenario directly: a hub sources from *N* edge/leafnode streams, then one chatty edge sources a deep 100k tail that buries the other edges' last messages. Crucially the hub store is **entirely sourced edge subjects** — as a real fan-in is — so every block matches the recovery sublist and *none* can be skipped. This is the work a freshly elected leader runs (`startingSequenceForSources`) before it can resume sourcing. One op == one full recovery.



edges	Before (reverse scan)	After (phase 1)	Δ
8	7.27 ms ±96%	0.194 ms ±23%	**−97.3%**
32	7.96 ms ±6%	0.286 ms ±8%	**−96.4%**
128	13.2 ms ±6%	0.609 ms ±19%	**−95.4%**
512	71.1 ms ±22%	3.68 ms ±45%	**−94.8%**
geomean	**15.3 ms**	**0.594 ms**	**−96.1%**

(filestore, single-server, `-benchtime=50x -count=6`, `benchstat`; all rows p=0.002 (n=6). Both columns on the same shared host; "before" measured by splicing in the pre-optimization function from the parent commit. The host is noisy — note the ±96% spread on the edges=8 "before" outlier — but the per-row deltas and the −96.1% geomean are stable across runs.)

Two effects compound, and the table separates them:

- **Store-bound vs source-bound.** "Before" carries a ~10 ms floor *independent of edge count* — it must reverse-scan the whole 100k tail to reach the buried quiet edges. "After" is flat in tail depth and scales only with edge count (one `LoadLastMsg` index jump each).

- **The O(sources²) sublist rebuild.** The old Phase 2 rebuilt the whole sublist on *every* source it resolved (`refreshSublist` per `update`). At 512 sources that is ~260k inserts per recovery — the "before" jumps to ~70–100 ms and (in a separate `-benchmem` run) **49.8 MB / 674k allocs/op**. Phase 1 resolves these without ever building a sublist, so allocations drop to **2,572/op** — ~260× fewer at 512 edges.

> Caveat: this is a micro-benchmark of the recovery function, not a full Raft election — an actual election adds a fixed quorum/round-trip cost on top of *both* columns equally. An earlier draft buried the edges under *direct* (non-sourced) hub traffic; there the old per-block `fss` skipping rescued the scan (only ~5×), which is not representative of a pure fan-in. The committed benchmark uses an all-sourced tail so nothing is skippable.

Isolating the store-depth axis

`BenchmarkJetStreamSourceResumeFanInTailDepth` fixes the edge count (32) and varies only the buried tail depth. It makes the store-bound vs source-bound split explicit: the old scan grows linearly with depth while Phase 1 stays flat, so the speedup *grows* with how deep the store is.

tail depth	Before (reverse scan)	After (phase 1)	Speedup
50k	0.89 ms	0.30 ms	3.0×
100k	11.6 ms	0.97 ms	11.8×
200k	23.1 ms	1.13 ms	20.5×
400k	46.0 ms	1.11 ms	41.6×

"Before" doubles with the tail (11.6 → 23.1 → 46.0 ms) — textbook O(depth). "After" is flat at ~1 ms and its allocations stay at **~223/op** regardless of depth, versus ~3,400/op for the old scan. (The 50k "before" is sub-linear because at that size the buried edges still sit within the first couple of blocks the scan reaches; from 100k on, the linear walk dominates.)

At scale (10M), and vs. a persisted resume map

The **persisted-resume-map approach** (a separate effort; see §11) persists a `map[source]→lastSeq` and *reads* it on recovery instead of recomputing from the store. To compare on the same axis, `BenchmarkJetStreamSourceResumeDeepStore` seeds the hub store directly — real end-to-end sourcing can't build a 10M-message store — with one `JSStreamSource`-headered anchor per source plus a deep tail, and a built-in check verifies phase 1 resolves every source before timing. This pushes the reverse-scan baseline to 10M:

sources	tail	Before (reverse scan)	After (index recompute)	Speedup
16	100k	8.6 ms	0.265 ms	33×
16	1M	96.9 ms	0.249 ms	389×
16	10M	972 ms	0.265 ms	3,668×
64	10M	974 ms	1.05 ms	928×
256	10M	991 ms	3.08 ms	322×

Both axes confirm the model: the reverse scan is O(depth) (8.6 → 97 → 972 ms across 100k → 1M → 10M) and ~flat in source count (it walks the whole tail regardless — 972/974/991 ms for 16/64/256 at 10M), while the index recompute is flat in depth (~0.27 ms at 16 sources from 100k to 10M) and scales with source count (~12 µs/source).

Cross-checking against the persisted-resume-map approach's published numbers (measured on a different host with larger messages), at 16 sources / 10M tail — each row is one recovery strategy and the benchmark it was measured in:

recovery strategy	benchmark	recovery time
reverse scan (pre-change baseline)	persisted-resume-map benchmark	5.03 s
reverse scan (pre-change baseline)	`…DeepStore` (this proposal)	0.972 s
index recompute (this proposal)	`…DeepStore` (this proposal)	0.265 ms
persisted resume map	persisted-resume-map benchmark	2.08 µs

Both reverse-scan figures have the same O(depth) shape; the ~5× absolute gap between the two benchmarks is environment (message size / host). The persisted resume map is ~100× faster than the index recompute *in absolute terms* because it never touches the store — a pure in-memory read — whereas the index recompute still loads one block per source. Those two factors compose to explain why the persisted-map benchmark's headline multiplier (~2.4M×) dwarfs the index-recompute multiplier (~3,700×): 2.42M / 3,668 ≈ 660 ≈ 5.2 (baseline gap) × 127 (map-read vs one-block-load). The two approaches are **complementary, not competing** — see §11.

What still forces a full `>` scan in Phase 2 (by design)

After the above, Phase 2 only widens to `>` (no block skipping) when a source *genuinely* needs it:

- a **catch-all source** — empty `FilterSubject`, no transform — that sources every subject of its origin (it really can land anywhere, so a broad match is correct); and

- an **exotic transform** (`partition` / `split` / `slice` /…) whose rendered destination is not a subject wildcard, so `transformUntokenize` can't reduce it to a matchable pattern.

Both are uncommon for the edge→hub fan-in. Everything else — distinct subjects, wildcard filters, `wildcard()` / `$N` transforms — either resolves in Phase 1 or narrows the Phase 2 sublist to a concrete or wildcard subject.

`BenchmarkJetStreamSourceResumePartitionFallback` measures this residual path with a `partition()` transform source (stored subjects `p.<n>` , which can't be reduced to a wildcard) alongside several distinct edges (always resolved in Phase 1), over a deep 100k all-sourced store:

placement of the `>` source	recovery	note
active (it's the chatty aggregator, last msg at LastSeq)	0.65 ms	the common case — the `>` scan resolves in O(1) despite the fallback
buried (quiet early, a distinct edge forms the tail)	56.2 ms	residual worst case — the `>` scan must walk the whole tail

The takeaway: the by-design `>` fallback is cheap whenever the catch-all/partition source is *active* (which is the normal state for an aggregator). It only degrades to a full scan when such a source is both present **and** has gone quiet under a deep tail — genuinely unavoidable, since a catch-all source has no single subject to index by. The distinct edges in the same stream are unaffected either way (Phase 1).

Testing plan & coverage

The recovery path has two layers worth testing separately: the **resolver** (`startingSequenceForSources` producing the right `si.sseq`) and the **end-to-end behaviour** (a real restart/election resumes sourcing exactly-once). Below is the current coverage and the prioritized gaps.

Layer 1 — resolver correctness (unit, calls the resolver directly). ✅ in place: `…IndexFastPath` (phase 1 + transform phase 2), `…Ambiguity` (catch-all + shared concrete destination), `…SetStartingSequenceForSourcesIndex` (update-path twin), `…TemplatedTransform` , `…SharedWildcardTransform` .

Layer 2 — end-to-end recovery. Added: `TestJetStreamSourcingResumeAfterRolloutRestart` — sources across both phases, hard-restarts the server (rolling-upgrade style), publishes a second batch, and asserts each origin's sourced sequences are exactly `1..N` (no gap = no missed resume; no duplicate = no re-sourced run). `TestJetStreamClusterSourcingResumeAfterLeaderStepDown` — the R3 equivalent:

steps down the agg leader so a new node runs the resolver, then asserts the same exactly-once property on the new leader. Both pass under `-race`.

Layer 1.5 — differential / property. Added: `TestJetStreamStartingSequenceForSourcesDifferential` — randomized layouts (mixed source kinds, random counts incl. zero, randomly interleaved depths) asserting the resolver's per-source `sseq` equals an independent brute-force scan of the same store, over a set of fixed reported seeds.

Prioritized gaps:

Pri	Area	Proposed test(s)	Guards against
~~P0~~ ✅	Clustered resume (R3)	`…ResumeAfterLeaderStepDown` (added)	resolver runs per-node on every election
~~P0~~ ✅	Property / differential	`…Differential` (added)	resolver regressions the hand-written cases miss
~~P1~~ ✅	Store-state edges (seeded)	`…SeededEdges` (added): pre-2.10 headers, subject overlap, interior delete of a source's last message	the header-`iname` disambiguation, pre-2.10 name-match, and `loadLast`'s deleted-skip
~~P1~~ ✅	`memstore` path	`…MemStore` (added)	the linear `LoadPrevMsgMulti` / index-light `LoadLastMsg` path for sources
~~P1~~ ✅	Exotic transforms	`…PartitionTransform` (added)	the residual Tier 2 `>` path resolves to the right seq
~~P2~~ ✅	Config-update path	`…ConfigUpdateScoping` (added): add/remove sources, plus a direct assertion that only the given inames are recomputed	`setStartingSequenceForSources` scoping (the `needsStartingSeqNum` path)
~~P2~~ ✅	First-seq > 1	`…FirstSeqAdvanced` (added): purge-by-sequence advances `FirstSeq`, both phases still resume	off-by-one in the reverse-scan / `loadLast` termination
P2 (covered)	Snapshot restore	shares the on-disk recovery path with `…ResumeAfterRolloutRestart`; a dedicated backup/restore round-trip remains optional	the cold-restore branch of recovery

Tier 0 (gated on implementation). When the persisted map lands, add a watermark matrix: `upToSeq == LastSeq` → trust; `upToSeq < LastSeq` → recompute (Tier 1); `upToSeq > LastSeq` → distrust; plus checksum-corruption and version-mismatch → fall through (degrade, not error); a crash between message durability and map flush → stale → Tier 1; clean `Stop` → Tier 0 hit; and a cluster snapshot round-trip.

Still to do

- All P0–P2 recovery-test gaps are covered and green under `-race`. The only remaining additions are the **Tier 0 watermark matrix** (gated on that feature landing) and, optionally, a dedicated stream backup/restore round-trip (the resolver path itself is already exercised by the restart test).

Resolved finding (was suspected pre-existing bug)

An earlier note suspected `setStartingSequenceForSources` mishandled transform sources because its Phase 2 sublist used `si.sfs` (transform **source** filters) rather than the **destination**. On inspection this was **not** a correctness bug: transform sources have an empty `FilterSubject`, so the old sublist inserted a `>` catch-all that matched the destination anyway (verified by `TestJetStreamStartingSequenceForSourcesTemplatedTransform`). It was purely an **efficiency** problem — the `>` defeated block skipping — now fixed by building the Phase 2 sublist from the destination wildcard form and dropping the dead `si.sfs` inserts.

11. Relationship to the durable-consumer proposal

This change is **orthogonal and complementary** to the durable-consumer/`si.sseq`-persistence ideas in `sourcing-durable-resume.md`:

- Durable consumers / persisted `si.sseq` aim to **skip** resume work entirely (when state survives).
- This proposal makes the **fallback resume itself cheap** — which still runs on cold start, on snapshot restore, for limits/clustered streams, and any time persisted state is missing.

The two were measured head-to-head at 16 sources / 10M tail (see "At scale" above): a persisted resume map reads in ~2 µs (pure in-memory, never touches the store), while the index recompute resolves in ~0.27 ms (one block load per source). Both are flat in store depth — versus ~1 s for the old reverse scan. The persisted map is the faster steady-state path, but it buys that speed with a write-path cost and a **new crash-consistency invariant**: the map must never lead durably-stored messages across truncation, compaction, message deletion, snapshot restore, and follower replication, or recovery resumes at a wrong sequence (gaps/duplicates). The index recompute carries none of that — the log plus the existing subject index remain the only source of truth — so it is the natural path to run whenever the persisted map is absent (pre-feature streams, cold start, restore) or must be revalidated.

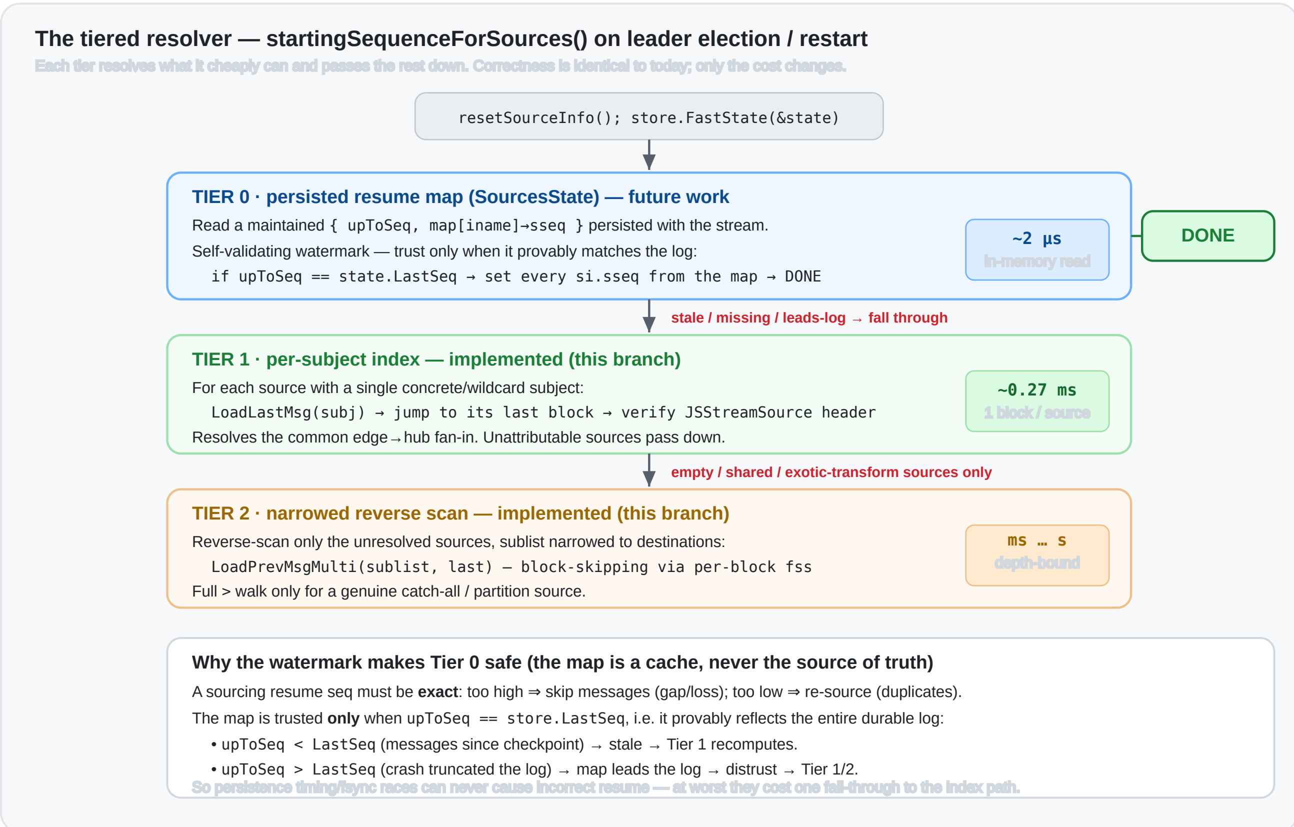
Recommended order: land this index-based resume first (self-contained, no protocol/state change, helps every retention type today, and replaces the O(depth) scan in the fallback the persisted design will still need), then layer the persistence/durable improvements on top.

12. Design sketch: Tier 0, a persisted resume map

This section sketches how the persisted-map approach (a parallel effort) would slot in as **Tier 0** on top of the resolver this branch already implements, what it buys, and the options/trade-offs for *where* the map lives. Nothing here is implemented on this branch — it is the design we'd build the two efforts toward.

12.1 The tiered resolver

`startingSequenceForSources` becomes three tiers, each resolving what it cheaply can and passing the remainder down. Correctness is identical at every tier — only the cost changes.



Tier	Mechanism	Resolves	Cost (16 src / 10M)	Status
0	read a persisted `{upToSeq, map[iname]→sseq}`	everything, if the map matches the log	~2 µs	design
1	`LoadLastMsg(subj)` per source via the subject index	distinct concrete/wildcard-subject sources	~0.27 ms	**this branch**
2	narrowed `LoadPrevMsgMulti` reverse scan	residual catch-all / exotic-transform sources	ms … s	**this branch**

Tier 0 is a pure *fast path*: when present and trustworthy it returns immediately; otherwise the work falls through to the index recompute (Tier 1) and, for the genuine residue, the narrowed scan (Tier 2). Because Tiers 1–2 already exist and are cheap, Tier 0 can be added incrementally with no correctness risk to the fallback.

12.2 How Tier 0 works

The per-source resume sequence (`si.sseq` — the origin sequence of the last message sourced from each source) is already tracked in memory: it is set in `processInboundSourceMsg` (`stream.go`) as each sourced message is ingested, and today it is **always recomputed** from the store on leader election (`setLeader` → `setupSourceConsumers` → `startingSequenceForSources`). No source/mirror resume state is persisted anywhere today.

Tier 0 persists that in-memory map so recovery can *read* it instead of recomputing:

1. **Maintain** `map[iname]→sseq` — already done; it is the set of `si.sseq` values.
2. **Persist** the map, tagged with a watermark `upToSeq` = the hub stream `LastSeq` the map reflects.
3. **On recovery**, load the map and decide whether to trust it (§12.4); on a hit, set every `si.sseq` from the map and return — no store access at all.

12.3 The consistency requirement (why a naive map is unsafe)

A sourcing resume sequence must be **exact**, not merely "not ahead":

- **Map too high** (claims a seq the store doesn't durably have — e.g. a crash truncated messages written after the last checkpoint) ⇒ we resume *past* real messages ⇒ **gap / data loss**.
- **Map too low** (stale — messages were sourced since the last checkpoint) ⇒ we re-request from the origin and re-append ⇒ **duplicates** (sourced messages carry no `Nats-Msg-Id` , so the store's dedupe does not catch them).

So "persist it and trust it" is only safe if the map is *exactly* the store's state — which is precisely the crash-consistency burden a persisted map introduces, and why the state is recomputed today.

12.4 The watermark: self-validating trust

The cheap way to get exactness without per-source verification (which would just re-do Tier 1) is a single **watermark**. Persist `upToSeq` alongside the map and, on recovery, compare it to the store's actual `LastSeq`:

Condition	Meaning	Action
`upToSeq == LastSeq`	map provably reflects the entire durable log	**trust it** — Tier 0 hit (~µs)
`upToSeq < LastSeq`	messages were stored after the checkpoint	stale → fall to Tier 1
`upToSeq > LastSeq`	crash truncated the log below the checkpoint	map leads the log → distrust → Tier 1/2

This makes Tier 0 **safe by construction and independent of fsync timing**: if the last few sourced messages weren't durable and are lost on restart, the recovered `LastSeq` is below `upToSeq`, the map is distrusted, and Tier 1 recomputes. The worst case of *any* persistence race is one fall-through to the index path — never an incorrect resume. The common clean-shutdown / steady-checkpoint case (the typical restart and rolling-upgrade path) hits `upToSeq == LastSeq` and pays ~µs.

12.5 Where to persist the map — options

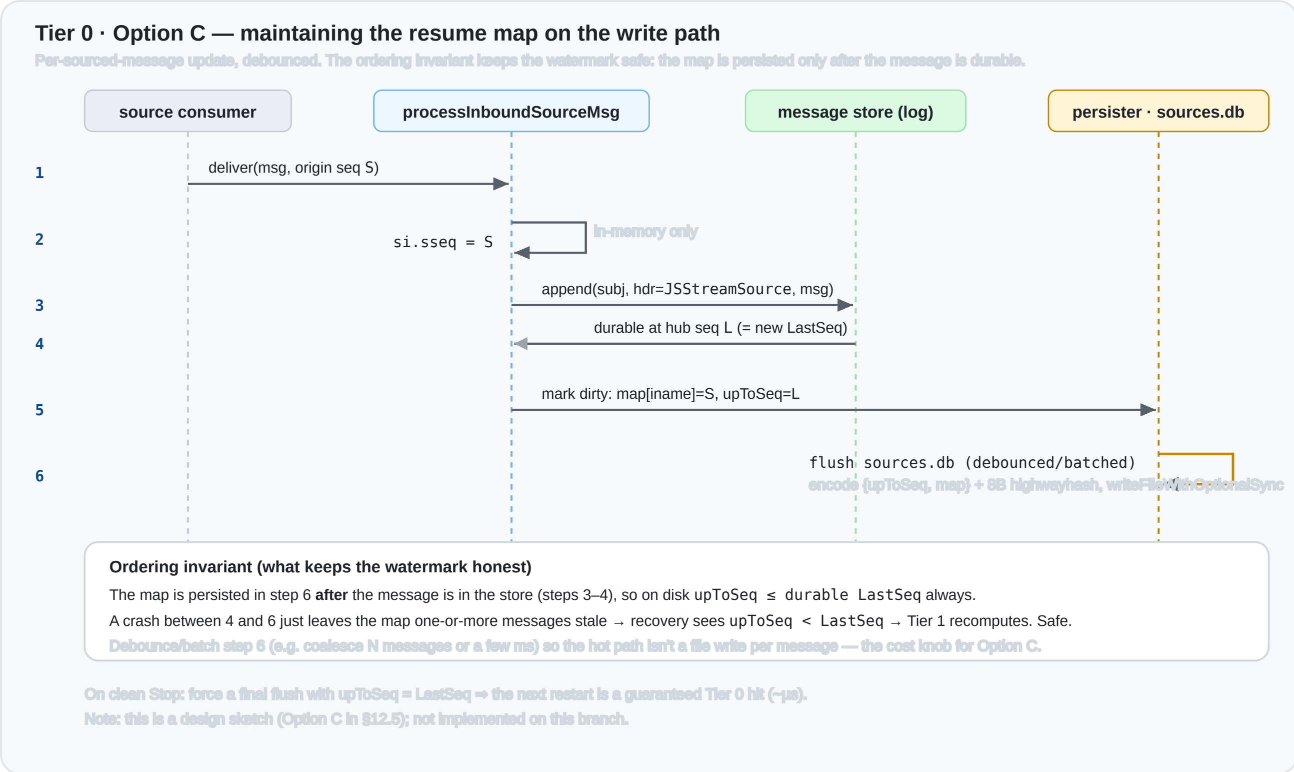
The map is tiny (`iname` → `uint64` , a few dozen entries), so the question is purely *where* it is written to stay consistent and *how often*. Five options, roughly increasing in consistency strength and in write-path / conflict cost:

Option	Mechanism	Pros	Cons
A. Inline in `index.db`	encode the map inside the filestore full-state file (`_writeFullState` / `recoverFullState`)	atomic with stream state; one file; watermark = the state's `LastSeq` for free	touches the checksummed filestore format (highest conflict surface); only as fresh as the periodic full-state write
B. Sidecar `sources.db`	a small separate file in the stream's `msgs/` dir, mirroring the	self-contained; no change to	a second file to write/sync and keep in

Option	Mechanism	Pros	Cons
	consumer-state file pattern (`encodeConsumerState` / `writeState` / highwayhash sum)	`index.db`; easy to version & checksum	step; its own staleness window
C. Write-path update	update + persist the map on every sourced message (the "always ~2 µs on recovery" design)	map is never stale (`upToSeq` ≈ `LastSeq` almost always) ⇒ Tier 0 hit even after a crash	per-message write amplification; needs batching/debounce; strongest crash-consistency coupling
D. Periodic + on-stop	snapshot `{LastSeq, map}` on a timer and on clean `Stop`	trivial; no write-path cost; on-stop write makes clean restarts a guaranteed Tier 0 hit	after a *crash*, almost always stale ⇒ falls to Tier 1 (still ~0.27 ms, so acceptable)
E. RAFT snapshot (clustered)	add the map to the stream's replicated snapshot encoding (`StreamReplicatedState` / `stateSnapshot`)	replicated to followers; survives leader change without a recompute	changes the replicated state format & version; must stay consistent across catchup/restore

Notes:

- **A vs B vs D differ only in the staleness window**, and the watermark makes any window *safe* — it only affects the Tier 0 *hit rate*, not correctness. So a low-risk first cut is **D** (periodic + on-stop, sidecar file), upgrading to **C** later if steady-state recovery latency on crash matters.
- **C** is what yields the headline "~2 µs even after a crash," at the cost of touching the hot write path; it is the most invasive and the one most worth measuring for write-amplification. Its update sequence and the ordering invariant that keeps the watermark honest are shown below:



- **E** is orthogonal to A–D and is required for the *clustered* win — otherwise a clustered stream still recomputes (now via Tier 1) when a new leader has no local map. This is the largest piece and the one with the most format/version care.
- Whatever the choice, the map must be **bounded by `LastSeq` on read** (§12.4) and treated as a hint the log can always override.

12.6 Clustered streams

For replicated streams the resume state lives behind RAFT. Two sub-cases:

- **Same node re-elected / restarted** — a local sidecar/ `index.db` map (A/B/D) applies as in the single-server case.
- **Different node becomes leader** — this is the *leader-election failover* case, and it is the one only option E helps. The new leader was a **follower**: it never ran the source consumers and never wrote a local map, so the local-file options (A/B/C/D) give it nothing — its watermark check can't pass and it falls through to Tier 1. Tier 1 keeps that fall-through cheap (~0.27 ms vs the old O(depth) scan), which is what makes Tier 0 *optional per node* rather than a hard dependency — but to get the ~µs Tier 0 hit on a freshly-elected node, the map has to be **replicated**, which is option E.

What option E involves

The replicated resume state rides the same RAFT machinery streams already use for catchup. The pieces:

1. **Carry the map in the replicated state.** Add a `Sources map[string]uint64` (iname → sseq) to `StreamReplicatedState` (`store.go:229`). The watermark is free: the snapshot is taken at a known `LastSeq` , which already lives in the same struct — so "map reflects the log up to `LastSeq` " needs no new field, just the §12.4 comparison on read.

2. **Encode/decode it.** Two strategies, trading conflict surface against simplicity:

   - **E1 — extend the binary stream-state format.** Append the map in `EncodedStreamState` (`filestore.go:12281` , `memstore.go:2369`) and read it in `DecodeStreamState` (`store.go:245`), behind a bumped `streamStateVersion` . Cleanest single blob, but touches the shared, version-checked stream-state codec (same high-conflict surface as option A — coordinate with whoever owns it).
   - **E2 — stream-layer envelope.** Leave the store codec untouched and have the stream append its own `SourcesState` block (the §12.9 encoding, reused verbatim) after the store's encoded state in `stateSnapshotLocked` (`jetstream_cluster.go:9905`), splitting it back off before `DecodeStreamState` on apply. Lower conflict surface; costs one extra length-prefix/parse step in the snapshot install/apply path.

3. **Populate on capture.** `stateSnapshotLocked` runs under the stream lock and already calls `EncodedStreamState` ; it would read the live `si.sseq` values out of `mset.sources` there (they are the map). No new bookkeeping — the values already exist in memory on the leader.

4. **Apply on the receiving node.** When a node installs/applies the snapshot (`processSnapshot` , `jetstream_cluster.go:10264` , and the apply path that decodes it), stash the decoded map on the stream (e.g. `mset.snapSources`). Then Tier 0 in `startingSequenceForSources` reads that map and applies it **iff** its watermark `== state.LastSeq` , exactly as in §12.4.

5. **Keep it fresh past the snapshot (the real failover nuance).** A snapshot is point-in-time. After an election the new leader typically *catches up* by applying log entries written after the snapshot — advancing `LastSeq` beyond the snapshot's watermark, so a plain snapshot map is stale and Tier 0 misses (falling to Tier 1). To actually hit Tier 0 on failover, the map must also be advanced as those entries apply: when an applied entry is a sourced message (it carries the `JSStreamSource` header), bump `snapSources[iname] = sseq` . That is option **C**'s update, but driven off the **apply** stream rather than the live ingest path — so every replica maintains the map for free as it applies, and any one of them can become leader with a current map.

6. **Compatibility.** Version-gate both directions: a node reading a snapshot without the map (older peer, or first rollout) simply finds no `Sources` and falls through to Tier 1; a node that doesn't understand the new field must skip it cleanly. Because the watermark already

makes a missing/stale map safe, mixed-version clusters degrade to the index recompute rather than mis-resuming — no flag-day.

Net: E is the largest and most consistency-sensitive option (it changes replicated state and rides the catchup/apply path), but it is also the only one that makes a **freshly-elected leader** resume in ~µs. Steps 1–4 give Tier 0 on a clean leader hand-off; step 5 is what extends it to a genuine failover with catchup. Throughout, Tier 1 remains the fallback, so a missing, stale, or version-mismatched map only ever costs a ~0.27 ms recompute — never a wrong resume.

E2 envelope — concrete sketch

The E2 strategy keeps the store's stream-state codec (`EncodedStreamState` / `DecodeStreamState`) untouched and wraps it. The wrinkle is backward compatibility: `DecodeStreamState` consumes its buffer to the end (the trailing delete-blocks have no length), so a `SourcesState` block can't simply be appended after a plain stream state — an old decoder would misread it as delete-blocks. The envelope therefore uses its own magic and a length prefix, and is **only emitted when every peer understands it** (gated exactly like the existing `supportsBinarySnapshotLocked`, `jetstream_cluster.go:9865`). Old peers keep receiving plain stream state and fall to Tier 1.

Byte layout (`snapshotEnvelopeMagic` is distinct from `streamStateMagic = 42` so the two are self-distinguishing on read):

Offset	Field	Type	Notes
`0`	`envMagic`	`u8`	`snapshotEnvelopeMagic` (≠ 42) — marks an enveloped snapshot
`1`	`envVersion`	`u8`	`= 1`
`2…`	`stateLen`	`uvarint`	byte length of the embedded stream state
`…`	`streamState`	`stateLen` bytes	verbatim `EncodedStreamState()` output (its own magic/version inside)
`…`	`sourcesBlock`	bytes	the §12.9 `SourcesState` encoding (its own header + highwayhash); `upToSeq` set to the embedded state's `LastSeq`

Wrap on capture (in `stateSnapshotLocked`, `jetstream_cluster.go:9905`):

```
func (mset *stream) stateSnapshotLocked() []byte {
	storeState, err := mset.store.EncodedStreamState(mset.getCLFS())
	if err != nil {
		return nil
```

```
	}
	// Only emit the envelope if the whole group understands it; else legac
	if !mset.supportsSourcesSnapshotLocked() {
		return storeState
	}
	var st StreamState
	mset.store.FastState(&st) // LastSeq is the watermark for the map below
	seqs := make(map[string]uint64, len(mset.sources))
	for iname, si := range mset.sources {
		if si.sseq > 0 {
			seqs[iname] = si.sseq
		}
	}
	src := encodeSourcesState(st.LastSeq, seqs, mset.store.hh()) // §12.9 e

	buf := make([]byte, 0, hdrLen+binary.MaxVarintLen64+len(storeState)+len
	buf = append(buf, snapshotEnvelopeMagic, snapshotEnvelopeVersion)
	buf = binary.AppendUvarint(buf, uint64(len(storeState)))
	buf = append(buf, storeState...)
	buf = append(buf, src...)
	return buf
}
```

Unwrap on apply (one helper the install/apply path calls instead of `DecodeStreamState` directly):

```
func decodeStreamSnapshot(buf []byte, hh hash.Hash64) (*StreamReplicatedState, m
	// Legacy / non-enveloped: a plain stream state, no sources map.
	if len(buf) < hdrLen || buf[0] != snapshotEnvelopeMagic || buf[1] != sn
		st, err := DecodeStreamState(buf)
		return st, nil, err
	}
	n := hdrLen
	stateLen, c := binary.Uvarint(buf[n:])
	n += c
	if c <= 0 || n+int(stateLen) > len(buf) {
		return nil, nil, ErrBadStreamStateEncoding
	}
	st, err := DecodeStreamState(buf[n : n+int(stateLen)])
	if err != nil {
		return nil, nil, err
	}
```

```
        upTo, seqs, err := decodeSourcesState(buf[n+int(stateLen):], hh) // §12
        if err != nil || upTo != st.LastSeq {
                // Tolerate a bad/skewed map: the snapshot still applies; resu
                return st, nil, nil
        }
        return st, seqs, nil
}
```

The apply path stashes the returned `seqs` on the stream (e.g. `mset.snapSources`) for Tier 0 to consult, and `DecodeStreamState` itself never changes — so a node that doesn't know the envelope (it shipped before the gate would let the leader emit one) is never handed these bytes in the first place. Two cheap invariants make it safe by construction: the distinct `envMagic` makes old vs new self-distinguishing, and `upTo != st.LastSeq` (or any decode error) drops the map and defers to the index recompute rather than trusting a skewed one.

12.7 Expected performance

From the measured numbers (§10, "At scale"), with 16 sources over a 10M-message store:

Path	Recovery	Scales with	Flat in store depth?
Old reverse scan (pre-branch)	~0.97 s	store depth	no — O(depth)
Tier 1 index recompute (this branch)	~0.27 ms	source count (~12 µs/src)	yes
Tier 0 persisted map (design)	~2 µs (reported)	source count (~0.13 µs/src)	yes

So Tier 0 is ~**100× faster than Tier 1 in absolute terms** (a pure in-memory read vs one block load per source) and both are flat in depth. The practical question is whether ~0.27 ms recovery is already good enough: for most deployments it is, and Tier 1 alone removes the multi-second cliff. Tier 0 earns its keep when recovery happens *often* or with *many* sources — frequent leader elections / rolling restarts, or hundreds–thousands of sources where 0.27 ms → tens of ms (Tier 1) is worth driving back to µs. It does **not** change the asymptotics (both are O(sources)); it lowers the constant by removing store I/O.

12.8 Recommendation

1. **Ship Tiers 1–2 now** (this branch): self-contained, no new state, removes the O(depth) cliff for every retention type, and is the fallback Tier 0 needs anyway.

2. **Add Tier 0 incrementally**: start with option **D** (periodic + on-stop sidecar, watermark-validated) for single-server/R1 — small, safe, and already turns clean restarts into µs. Measure the write path before considering **C**.
3. **Then Tier 0 for clusters** (option **E**): the largest and most consistency-sensitive piece; gate it behind the same watermark and keep Tier 1 as the per-node fallback so it can never resume incorrectly.

12.9 Concrete encoding (`SourcesState`)

The map is small, so a flat varint encoding mirroring `encodeConsumerState` (`store.go:401`) is plenty. It reuses the filestore conventions: a 2-byte header (`magic = 22` , `version = 1` , `hdrLen = 2` , `filestore.go:288–294`) and an 8-byte trailing highwayhash-64 checksum over the preceding bytes (`checksumSize = 8` , computed with the store's `fs.hh` digest as in `_writeFullState`).

Byte layout:

Offset	Field	Type	Notes
`0`	`magic`	`u8`	`= 22` (filestore magic)
`1`	`version`	`u8`	`= 1`
`2…`	`upToSeq`	`uvarint`	hub `LastSeq` the map reflects — the watermark (§12.4)
`…`	`n`	`uvarint`	number of source entries
`…`	`len(iname)`	`uvarint`	┐ repeated `n` times
`…`	`iname`	`bytes`	│ the source's unique index name
`…`	`sseq`	`uvarint`	┘ last origin sequence sourced for it
end−8	`checksum`	`[8]byte`	highwayhash-64 over `buf[0 : end−8]`

Encode (sketch):

```
const sourcesStateMagic = magic // 22, shared filestore magic
const sourcesStateVersion = uint8(1)

// encodeSourcesState serializes {upToSeq, map[iname]→sseq}. hh is the store's
// keyed highwayhash digest (fs.hh); pass nil to skip the checksum (tests).
func encodeSourcesState(upToSeq uint64, seqs map[string]uint64, hh hash.Hash64)
        // Upper bound: header + upToSeq + count + per-entry(len + iname + ssec
        sz := hdrLen + 2*binary.MaxVarintLen64
```

```
	for iname := range seqs {
		sz += binary.MaxVarintLen64 + len(iname) + binary.MaxVarintLen
	}
	sz += checksumSize
	buf := make([]byte, sz)

	buf[0], buf[1] = sourcesStateMagic, sourcesStateVersion
	n := hdrLen
	n += binary.PutUvarint(buf[n:], upToSeq)
	n += binary.PutUvarint(buf[n:], uint64(len(seqs)))
	for iname, sseq := range seqs {
		n += binary.PutUvarint(buf[n:], uint64(len(iname)))
		n += copy(buf[n:], iname)
		n += binary.PutUvarint(buf[n:], sseq)
	}
	if hh != nil {
		hh.Reset()
		hh.Write(buf[:n])
		n += copy(buf[n:], hh.Sum(nil)) // 8 bytes
	}
	return buf[:n]
}
```

Decode (sketch) — rejects a bad magic/version/checksum and returns the map plus the watermark, which `startingSequenceForSources` (Tier 0) then compares against `state.LastSeq`:

```
func decodeSourcesState(buf []byte, hh hash.Hash64) (upToSeq uint64, seqs map[st
	if len(buf) < hdrLen+checksumSize || buf[0] != sourcesStateMagic || buf
		return 0, nil, errBadSourcesState
	}
	body, sum := buf[:len(buf)-checksumSize], buf[len(buf)-checksumSize:]
	if hh != nil {
		hh.Reset()
		hh.Write(body)
		if !bytes.Equal(hh.Sum(nil), sum) {
			return 0, nil, errBadSourcesState
		}
	}
	n := hdrLen
	upToSeq, c := binary.Uvarint(body[n:]); n += c
```

```
        cnt, c := binary.Uvarint(body[n:]); n += c
        seqs = make(map[string]uint64, cnt)
        for i := uint64(0); i < cnt; i++ {
                l, c := binary.Uvarint(body[n:]); n += c
                iname := string(body[n : n+int(l)]); n += int(l)
                sseq, c := binary.Uvarint(body[n:]); n += c
                seqs[iname] = sseq
        }
        return upToSeq, seqs, nil
}
```

Size: ~`len(iname)+~12` bytes per source, so a 256-source stream is well under 10 KB — a single small write, well within one filesystem block for typical fan-ins. Forward compatibility is the usual `version` bump (Tier 0 simply distrusts an unrecognized version and falls through to Tier 1, so an old/new format mismatch degrades to the index recompute rather than an error).

The Tier 0 read in `startingSequenceForSources` is then small:

```
// Tier 0: trust a persisted map only if it provably matches the durable log.
if buf, _ := mset.readSourcesState(); len(buf) > 0 {
        if upToSeq, seqs, err := decodeSourcesState(buf, mset.store.hh()); err
                for iname, sseq := range seqs {
                        if si := mset.sources[iname]; si != nil {
                                si.sseq, si.dseq = sseq, 0
                        }
                }
                return // sources not in the map have never sourced → sseq sta
        }
}
// else fall through to Tier 1 (index recompute) / Tier 2 (narrowed scan)
```

Appendix — key references

Symbol	File:line	Role
`startingSequenceForSources`	`stream.go:4694`	the scan to replace (Phase 1 + Phase 2)

Symbol	File:line	Role
unconditional call	`stream.go:4821`	runs on every leader election / restart
`setStartingSequenceForSources`	`stream.go:4566`	update-path twin, same treatment
sublist build (stored subjects)	`stream.go:4742-4757`	defines a source's stored subject(s)
`LoadLastMsg` / `loadLast`	`filestore.go:9048` / `8939`	index-based last-msg-for-subject (returns header)
`MultiLastSeqs`	`filestore.go:3806`	batched last-seq-per-filter via index
`psi` / `fs.psim`	`filestore.go:168` / `195`	per-subject `{total, fblk, lblk}`; store-wide subject→blocks index
`mb.fss` / `SimpleState`	`filestore.go:239` / `store.go:180`	per-block subject→ `{Msgs, First, Last}` index
`psim` maintenance on write	`filestore.go:4933-4943`	how `total` / `lblk` are kept current
`LoadPrevMsgMulti`	`filestore.go:9403` / `memstore.go:1991`	the backward walk used by Phase 2
`Nats-Stream-Source` header	`stream.go:635`	constant; the per-message source provenance
`genSourceHeader` / `streamAndSeq`	`stream.go:4470` / `4545`	writes / parses origin stream, iname, origin seq

Tier 0 (design) persistence hooks:

Symbol	File:line	Role
`processInboundSourceMsg` (`si.sseq = sseq`)	`stream.go:4371`	where the in-memory resume map is updated per sourced message
`setLeader` → `subscribeToStream` → `setupSourceConsumers`	`stream.go:1255` / `4957` / `4927`	the recovery trigger Tier 0 would short-circuit

Symbol	File:line	Role
`_writeFullState` / `recoverFullState`	`filestore.go:11681` / `1871`	filestore full-state write/read (`index.db`) — option A host
`encodeConsumerState` / `writeState`	`store.go:401` / `filestore.go:13164`	consumer-state file pattern to mirror for a sidecar `sources.db` (option B)
`StreamReplicatedState` / `stateSnapshot`	`store.go:229` / `jetstream_cluster.go:9897`	replicated stream state / snapshot — option E host (no source state today)
`EncodedStreamState` / `DecodeStreamState`	`filestore.go:12281` , `memstore.go:2369` / `store.go:245`	replicated-state codec to extend for option E1 (version-gated)
`stateSnapshotLocked` / `processSnapshot`	`jetstream_cluster.go:9905` / `10264`	snapshot capture (populate map) and apply (seed map + maintain from applied entries) for option E